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Combined pulmonary fibrosis and emphysema syndrome: clinical research progress

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Abstract
Combined pulmonary fibrosis and emphysema (CPFE) is a distinct radiological–clinical syndrome defined by the coexistence of upper-lobe emphysema and lower-lobe fibrosis. It predominantly affects elderly heavy smokers and exhibits a hallmark physiological paradox: preserved lung volumes with severely impaired gas exchange and exertional hypoxaemia. This review synthesises current evidence on CPFE pathogenesis, diagnosis, and management, focusing on challenges arising from its unique pathophysiology. We critically examine the diagnostic implications of the pulmonary function "dissociation phenomenon" and the limitations of quantitative imaging. Persistent ambiguity in diagnostic criteria, variable geographic recognition, and the practical applicability of current definitions are also addressed. Management remains largely extrapolated from idiopathic pulmonary fibrosis (IPF) and chronic obstructive pulmonary disease (COPD). High-quality, CPFE-specific evidence is lacking, and no dedicated therapeutic framework exists. This review identifies key research gaps and underscores the urgent need for prospective studies to establish robust diagnostic and treatment strategies for this complex syndrome.

Keywords Pulmonary Fibrosis, Pulmonary Emphysema, Combined Pulmonary Fibrosis and Emphysema, Complications and Outcome, Management, Treatment

Introduction

COPD and pulmonary fibrosis have long been recognized as distinct pulmonary disease entities, exhibiting substantial discrepancies in pathological mechanisms, imaging features, and clinical management strategies [1, 2]. With the continuous advancement of imaging techniques and the deepening of clinical investigations, accumulating evidence has demonstrated that emphysema and pulmonary fibrosis can coexist in a single patient, thereby forming an overlapping syndrome with unique clinical, physiological, and pathological profiles. In 2005, Cottin et al. first formally introduced the concept of CPFE [3]. The insidious pathological mechanisms, non-specific clinical presentations, and absence of standardized diagnostic criteria frequently result in delayed diagnosis of CPFE, which in turn impedes the implementation of effective clinical management. Importantly, significant geographical heterogeneity is observed in the epidemiology, etiology, and clinical phenotypes of CPFE. Against this background, this review is intended to systematically collate the core advances in basic and clinical research on CPFE, with the ultimate goal of optimizing its clinical diagnosis and treatment practices.

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Epidemiology

Reported prevalence rates of CPFE vary markedly. Prevalence estimates among patients with IPF range from 8 to 67%, while in cohorts with idiopathic interstitial pneumonia (IIP), reported rates vary between 26 and 54%. A particularly high prevalence (approximately 73%) has been documented in male patients with lung cancer. Geographically, higher prevalence rates are reported in regions such as Asia and Greece compared to the United States [4]. The epidemiological profile in Egypt presents a notable discrepancy. As a country with a high burden of tuberculosis and a concurrently elevated smoking prevalence, a higher prevalence of CPFE would be anticipated. However, reported rates remain markedly low, ranging from only 3.2% to 8.7% [5]. The pronounced divergence in prevalence estimates stems from fundamental heterogeneities in cohort selection, diagnostic standards, adjustment for confounders, and analytical approaches. To enable valid cross-population comparisons and enhance the clinical understanding of CPFE, future epidemiological research must prioritize the development and widespread adoption of international consensus-based diagnostic criteria.

Etiology and pathogenesis

The pathogenesis of CPFE syndrome is multifactorial and not fully elucidated. likely involving interactions among multiple pathways, including environmental exposures (notably tobacco smoke), genetic factors, and autoimmunity.

Smoking

Smoking is a major risk factor for CPFE with a well-demonstrated dose–response relationship [4, 6, 7]. Mejía et al. showed that both smoking frequency and duration were higher in IPF patients with emphysema [8]. Furthermore, KJ Chae et al.'s retrospective study of 2016 asymptomatic smokers found a 3.1% prevalence of CPFE in this cohort. Persistent smoking drives CPFE progression: 86% of patients who continued smoking post-diagnosis experienced disease progression [9].

The pathophysiology of smoking-induced lung injury in CPFE is multifaceted. Tobacco smoke, a complex aerosol containing numerous toxicants and oxidants, inflicts direct damage on the respiratory epithelium, alveolar structures, and pulmonary vasculature.Beyond direct cytotoxicity, smoke constituents dysregulate critical homeostatic systems—notably the oxidant/antioxidant and protease/antiprotease balances—and promote a pro-inflammatory, pro-fibrotic microenvironment by activating aberrant immune responses, altering macrophage polarization, and inducing epithelial cell apoptosis [10]. Furthermore, chronic smoke exposure drives epigenetic modifications. Collectively, these mechanisms contribute to the distinct "smoking-related interstitial lung disease" phenotype, clinically characterized by the frequent co-occurrence of IPF-like fibrosis, emphysema, radiographic honeycombing, and associated comorbidities such as coronary artery disease [11].

Heredity

Genetic factors are implicated. Guo et al. through bioinformatic analysis of genes set from the GEO database, identified a signature of 13 upregulated and 3 downregulated genes. Functional enrichment analysis indicated these genes are principally involved in immune regulation, extracellular matrix remodeling, and signal transduction pathways [12]. In another study, Tanwar et al. applied a random forest classifier combined with the SHAP algorithm to identified *ADRB2, CDH3*, and *IRS2* as key candidates, which are closely involved in airway function regulation, inflammatory response, and cellular apoptosis—biological processes tightly linked to CPFE pathogenesis [13]. Bioinformatics has generated promising hypotheses on the genetic and immune mechanisms of CPFE, but its clinical utility is limited. Current research focuses mainly on single-gene associations with airway function, ignoring CPFE's hallmark "fibrosis + emphysema" dual pathology. Additional limitations include unclear causality, inadequate exploration of genetic heterogeneity and gene-environment interactions, and insufficient external validation. Future research should prioritize independent cohort validation and integrate multi-omics with clinical phenotypes to establish causality and translation.

Autoimmunity

CPFE patients—regardless of elevated serum antinuclear antibody (ANA) or perinuclear anti-neutrophil cytoplasmic antibody (p-ANCA) status—exhibited significant differences compared to IPF patients without emphysema, according to a multicenter study. CPFE patients with a positive serum immune profile showed abundant *CD20 + B* cell infiltrates forming lymphoid follicles in fibrotic lung regions, a finding associated with improved survival outcomes [14]. Connective tissue diseases (CTDs)—notably systemic sclerosis and rheumatoid arthritis—are strongly associated with CPFE [15], consistent with observations by Bon San Koo et al. [16]. Collectively, these studies indicate that autoimmune pathways contribute to CPFE pathogenesis.

Clinical manifestations

CPFE predominantly affects elderly male smokers, with a mean age of 65–70 years. Its clinical manifestations are generally nonspecific, with exertional dyspnea, cough, and reduced exercise tolerance as primary features. Physical examination may reveal crackles and digital clubbing

[4, 17]. CPFE patients with pulmonary hypertension(PH) experience more severe dyspnea, and their cardiac function classifications according to the New York Heart Association tend to be Class III or IV [18]. Compared to IPF patients alone, CPFE patients have higher prevalence of cardiovascular and cerebrovascular disease, more frequent wheezing, and fewer crackles [19].

Pulmonary function and physiology

CPFE is characterized by distinct pulmonary function features, with relatively preserved lung volumes and markedly impaired diffusing capacity being the most prominent. In CPFE patients, the coexistence of fibrotic and emphysematous pathologies leads to a mutual offset of restrictive and obstructive ventilatory defects, which accounts for the near-normal values of FEV1/FVC, RV, and RV/TLC ratio. In sharp contrast, the diffusing capacity of the lung for carbon monoxide (DLCO) is significantly reduced due to alveolar dysfunction. This impairment is closely linked to the decreased pulmonary capillary volume resulting from fibrosis, as well as the ventilation-perfusion mismatch synergistically induced by both fibrotic and emphysematous changes [4, 20].

Notably, for individuals with normal routine respiratory parameters, the omission of DLCO testing may result in the underdiagnosis or underestimation of CPFE [21]. At present, no single physiological parameter has been validated as an optimal surrogate for monitoring disease progression. Hence, a composite approach integrating serial clinical assessment, HRCT, and multiparametric pulmonary function testing is mandated to accurately capture the complex trajectory of this dual-compartment disease [4, 17].

Imaging features and quantification

CPFE diagnosis relies primarily on HRCT findings. While emphysema and interstitial fibrosis can manifest in any lung region or subtype, characteristic HRCT features include upper-lobe emphysema and lower-lobe fibrosis [4, 22] (Fig. 1). Emphysema is defined as CT-detected low-attenuation regions without visible walls, and can be categorized as centrilobular, paraseptal, or panacinar, with paraseptal emphysema being the most common [3, 23, 24]. Usual interstitial pneumonia (UIP) is the predominant fibrotic pattern [3, 25].

Imaging is preferred over pulmonary function tests for assessing emphysema, as fibrosis can confound physiological measurements [4]. To date, no unified imaging scoring system for CPFE has been established; instead, most current evaluation approaches adapt the individual scoring criteria developed for emphysema and pulmonary fibrosis separately. Traditional visual assessment methods, such as the Goddard score for emphysema and the Camiciottoli score for fibrosis, are associated with several limitations, including substantial interobserver variability, time intensiveness, discrepancies in spatial resolution, and inadequate quantitative accuracy [4, 26–28]. Quantitative CT (qCT) analysis, as an objective tool for quantifying the extent of emphysema and fibrosis, has been validated for assessing disease severity and

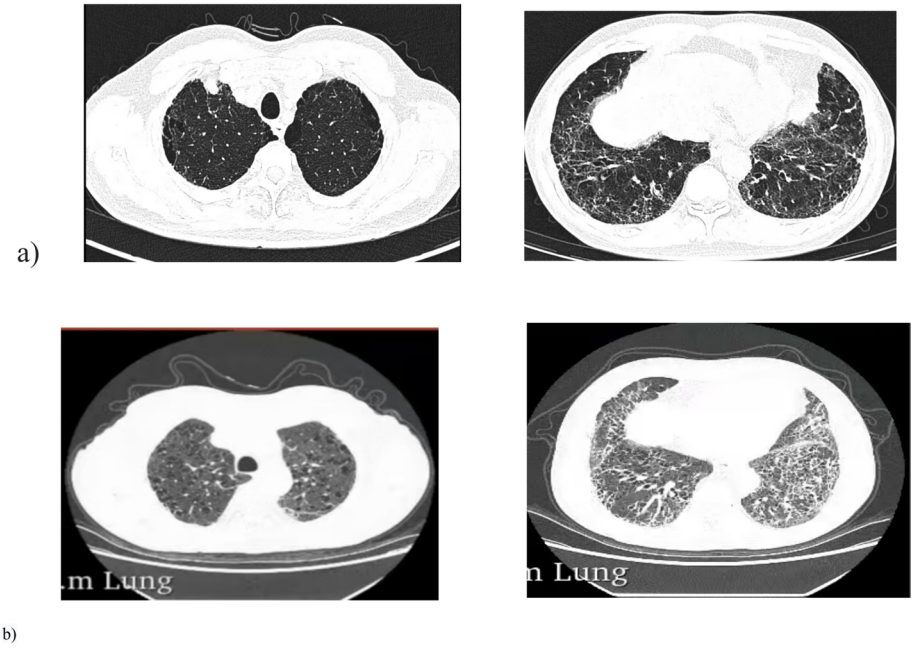


Fig. 1 Imaging of a typical case of CPFE: (**a**) the upper zones of the lungs showing predominant paraseptal emphysema, the lower zones of the lungs showing reticular opacities, honeycombing, and traction bronchiectasis. (**b**) the upper zones of the lungs showing predominant centrilobular lemphysema, the lower zones of the lungs showing reticular opacities, honeycombing, and traction bronchiectasis

predicting prognosis in patients with IPF and CPFE [29, 30]. However, qCT still faces considerable challenges: it is difficult to discriminate between low-attenuation areas of emphysema, honeycombing cysts, and low-attenuation changes related to traction bronchiectasis. Thus, until this diagnostic dilemma is resolved, visual quantification of emphysema remains the first-line approach [4]. Additionally, quantitative assessment of fibrosis is flawed; for example, lung volume loss induced by fibrosis and inflammatory changes misclassified as fibrosis can both compromise the accuracy of evaluations. The CPFE index [31], despite clinical value, lacks extensive external validation and anatomical specificity. Future advances should focus on standardizing qCT methodologies, developing advanced algorithms to enhance tissue differentiation, and establishing integrated assessment models that validate and combine imaging and functional indicators.

Diagnosis

A uniform definition of CPFE syndrome is currently lacking. To facilitate both future research and clinical practice, the 2022 ATS/ERS/JRS/ALAT statement proposes separate research and clinical definitions [4]. The research definition comprises: (1) coexistence of emphysema and pulmonary fibrosis on HRCT; (2) emphysema involving ≥ 5% of total lung volume; and (3) any subtype of pulmonary fibrosis. The clinical classification criteria require meeting the research definition plus one or more of the following: (i) emphysema extent ≥ 15% of total lung volume; (ii)relatively preserved lung volumes and airflow with disproportionately reduced DLCO, especially in patients with limited HRCT abnormalities and no pulmonary hypertension; and (iii) precapillary pulmonary hypertension not attributable to the sole presence of emphysema (FEV1 > 60%), fibrosis (FVC > 70%), or etiology (e.g., absence of CTD) [4, 17]. The 2022 ATS/ERS/JRS/ALAT statement have established an initial operational framework for the diagnosis of CPFE. However, significant limitations persist in terms of diagnostic accuracy and standardization, which are mainly reflected in the following two core aspects:diagnostic ambiguity and cognitive bias.

At the core of diagnostic uncertainty lies the difficulty in reliably discriminating true CPFE from other disease entities that share overlapping morphological features. Two distinct clinical scenarios epitomise this challenge: (i) the identification of mild emphysema co-existing with early fibrotic changes, and (ii) the differentiation of CPFE from advanced COPD with secondary fibrotic scarring. Mild emphysema is typically characterized by scattered, small non-walled low-attenuation areas in the lungs, which do not strictly conform to the classic upper lobe predominance. Early pulmonary fibrosis is mainly presented as mild interlobular septal thickening, subpleural micronodules, and focal ground-glass opacities, without the formation of typical UIP features such as honeycombing and traction bronchiectasis. Strict adherence to research-defined thresholds may lead to the exclusion of such patients; however, emerging evidence indicates that even mild fibrosis can impact prognosis, thereby challenging the rationality of using a single imaging volume threshold as the gold standard for CPFE diagnosis. In the long-term course of COPD, repeated airway inflammation and an imbalance in lung tissue injury-repair processes can lead to secondary intrapulmonary fibrotic scars. In contrast, fibrosis in CPFE represents the independent coexistence of IPF and emphysema, where the fibrotic component is mostly UIP-patterned and exhibits lower lobe predominance. Fibrotic scars secondary to COPD, however, are reparative changes following chronic inflammation of the airways and lung parenchyma, which are mostly focal and scattered, lack typical lower lobe predominance, and are always centered on airway obstructive pathology [1, 2, 4, 24]. Unfortunately, HRCT only allows visualization of the morphological features of fibrosis but cannot distinguish its pathological origin, which remains a key obstacle to the differential diagnosis of these two entities.

The recognition of CPFE as a discrete clinical syndrome remains uneven across geographical and healthcare settings. In the Egyptian context, for example, the concept is not yet widely assimilated into routine practice. A survey by Abdel-Mohsen et al. revealed that only 35.7% of pulmonologists could accurately define CPFE, reflecting a knowledge framework still oriented toward single-disease diagnostic paradigms [5]. This cognitive gap is compounded by disparities in resource availability and technical capacity. Primary and secondary care centres frequently depend on conventional CT or chest radiography, lack access to DLCO measurement, and provide insufficient training in the identification of subtle combined pathologies. Consequently, the very criteria intended to standardise diagnosis—dichotomous definitions, stringent exclusion conditions, and quantitative CT thresholds—become sources of inconsistency when implemented in settings where precise densitometric quantification is impractical. Thus, the current diagnostic framework inadvertently exacerbates, rather than alleviates, diagnostic heterogeneity (Fig. 2) (Table 1).

Complications

Pulmonary hypertension

PH complicates 15–55% of CPFE cases and confers a poor prognosis [4]. CPFE-associated PH is classified as WHO Group III [3]. Right heart catheterisation is the diagnostic gold standard but is invasive; most studies therefore use echocardiography (systolic pulmonary arterial pressure > 45 mmHg) [3]. Meta-analyses confirm that

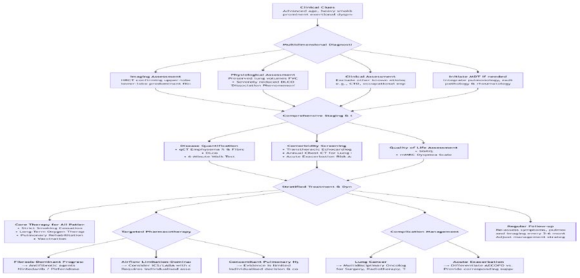


Fig. 2 Clinical management pathway for CPFE

PH is more prevalent and severe in CPFE and independently predicts mortality—one-year mortality reaches 60% in non-transplanted patients [32, 33]. Meta-analyses confirm that PH is more prevalent and severe in CPFE and independently predicts mortality—one-year mortality reaches 60% in non-transplanted patients. Cottin et al.'s study of 40 CPFE patients found right heart catheterization measured a mean pulmonary artery pressure of 40 ± 9 mmHg, indicating moderate to severe hemodynamic impairment [34]. Mejía et al.'s comparative research demonstrated that CPFE patients have more severe PH and higher mortality than those with fibrosis alone [8].

The pathophysiology of PH in CPFE is multifactorial, including synergistic pulmonary capillary bed destruction by emphysema and fibrosis, chronic hypoxia, and increased vascular resistance. Mounting evidence indicates cigarette smoke directly induces vascular dysfunction, inflammation, oxidative stress, and a procoagulant state, driving pathological vascular remodeling and PH progression [35].

Acute exacerbation

Acute exacerbation of IPF(AEIPF) occurs in CPFE, defined as acute dyspnoea worsening,within 1 month, new bilateral infiltrates, and exclusion of alternative causes. Annual incidence is 9.4%, with high rates of respiratory failure and poor outcomes [36, 37]. Ikuyama et al.'s retrospective study showed longer post-exacerbation survival in CPFE than in IPF alone [38]. A qCT study found that CPFE patients with ≥ 10% fibrosis had the poorest overall survival and highest AEPF incidence [39]. Zantah et al. classified AE-CPFE into AECOPD and AEIPF,noting AEIPF more frequently requires mechanical ventilation or ECMO [40], highlighting the prognostic and therapeutic relevance of exacerbation subtyping.

Lung cancer

The incidence of lung cancer in CPFE patients is 2%–52%, and typically localises to fibrotic zones [4, 41]. Adenocarcinoma is the predominant histological subtype, followed by squamous cell carcinoma [41]. In contrast, Feng et al. reported adenocarcinoma as the most common

Table 1 Clinical differentiation between CPFE and advanced COPD with fibrosis

Diagnostic Dimension	Combined Pulmonary Fibrosis and Emphysema (CPFE)	Advanced COPD with Concomitant Fibrotic Scarring
Core Pathology & Distribution	Coexistence of two independent pathologies: • Emphysema: Predominantly paraseptal or centrilobular in the upper lobes • Fibrosis: Predominantly a usual interstitial pneumonia (UIP) pattern in the lower lobes, with subpleural and basal predominance	Fibrosis as a secondary phenomenon of airway disease: • Emphysema: Typically diffuse centrilobular • Fibrosis: Manifests as focal, irregular scars or fibrous bands, often adjacent to emphysematous areas or along bronchovascular bundles. Lacks the classic lower-lobe, UIP-pattern distribution
Physiological Characteristics	Characteristic "dissociation phenomenon: • Preserved FEV1/FVC ratio and lung volumes (e.g., FVC) • Severely and disproportionately reduced DLCO (often < 40% predicted)	Predominant obstructive ventilatory defect: • Markedly reduced FEV1/FVC ratio • Lung hyperinflation (increased RV/TLC) • DLCO may be reduced, but its impairment typically correlates with the severity of airflow limitation (FEV1), lacking the distinct "dissociation" seen in CPFE
Clinical Presentation & Risk Profile	• Exertional dyspnea and hypoxemia are disproportionately severe relative to preserved lung volumes on imaging/exam • Digital clubbing is more common • Extremely strong association with smoking history	• Dyspnea severity aligns with COPD severity (e.g., GOLD grade) • Chronic cough and sputum production are typical • Digital clubbing is rare
HRCT Imaging Details	Asymmetric biphasic pattern: • Upper lobes: "Destructive" low-attenuation areas (emphysema) • Lower lobes: "Proliferative" reticulation, honeycombing (fibrosis) • The two patterns are spatially distinct	Intermingled, heterogeneous pattern: • Emphysema and fibrotic changes are diffusely admixed • Fibrosis appears as traction bronchiectasis, irregular linear opacities, or focal consolidation, not as extensive subpleural honeycombing in the lower lobes
Disease Course & Therapeutic Response	• Poorer prognosis, with a tendency for rapid progression • Mortality risk is often linked to fibrosis progression, pulmonary hypertension, or acute exacerbations • Poor response to bronchodilators alone	• Disease course is driven primarily by airflow limitation and COPD exacerbations • Typically shows partial reversible response to bronchodilators and inhaled corticosteroids

histological subtype of lung cancer in CPFE patients, followed by squamous cell carcinoma [42]. Compared to IPF-NSCLC, CPFE-NSCLC confers a significantly higher risk of acute exacerbation [43]. Importantly, acute exacerbation is the dominant predictor of mortality [44], underscoring that prognosis in CPFE-related lung cancer is determined more by the underlying disease than by cancer stage itself.

Prognosis and outcome

CPFE prognosis is poor: median survival 2.1–8.5 years, 5-year survival 38–55% [45]. Mortality predictors include low DLCO, age, and comorbidities [2]. Zhao et al. showed that in IPF with > 10% emphysema, annual DLCO decline outperforms FVC decline in predicting mortality [46]. Health-related quality of life is significantly impaired in CPFE. IPF-PRO data indicate that patients with CPFE report worse St. George's Respiratory Questionnaire (SGRQ) activity scores and more severe cough-related symptoms on the Cough and Sputum Assessment Questionnaire (CASA-Q) than in IPF alone [47]. Suzuki et al. identified quantitative CT parameter %LAA (a marker of emphysematous and fibrotic lesions) as an independent predictor of hospitalization in CPFE patients [48]. A retrospective study by the International Society of Heart and Lung Transplantation (ISHLT) documented favorable outcomes in CPFE lung transplant recipients: 1-,3-,and5-year survival rates of 94%,82%,and 74%, respectively, alongside preserved long-term graft function [33], highlighting lung transplantation's benefits, feasibility, and safety in suitable CPFE candidates.

Management and treatment

General measures

Smoking cessation, the cornerstone of CPFE treatment, and avoidance of other inhalational exposures directly target smoke-induced lung injury. Standardized oxygen therapy is fundamental; long-term home oxygen therapy benefits CPFE patients with resting or exercise-induced hypoxemia, relieving hypoxia, improving cardiopulmonary function, and reducing right heart burden, especially those with PH.

Pulmonary rehabilitation (breathing training, aerobic exercise, nutritional counseling, psychological support) improves CPFE patients' exercise tolerance, relieves dyspnea, and enhances quality of life. Routine vaccination against influenza, pneumococcus, and SARS-CoV-2 is recommended to prevent acute exacerbations [4, 49].

Drug treatment

Drug treatment for CPFE lacks evidence-based support. Current management is extrapolated from IPF and COPD [4] (Table 2). A small CPFE study on inhaled corticosteroids (ICS)/long-acting β2-agonists (LABA) reported improved lung function and reduced acute exacerbations [4, 50]. A prospective study by Wang et al. further showed ICS/LABA improved lung function, HRCT scores, and reduced pulmonary artery pressure in CPFE patients [51], providing preliminary evidence. but evidence is limited to small, non-randomised studies,unevaluated long-term outcomes. Antifibrotics (nintedanib, pirfenidone) slow FVC decline in IPF [52], but extrapolation to CPFE is problematic. Coexisting emphysema alters lung structure and may modify drug efficacy, dosing, and safety. The benefit–risk balance is uncertain.Thus,dedicated CPFE trials are urgently needed to inform reliable guidelines.

Treatment of complications

Lung cancer in CPFE follows standard oncologic management. Surgical resection is preferred when feasible; alternatives include chemotherapy, radiotherapy, targeted therapy, or palliation [4]. PH pharmacotherapies include NO-cGMP pathway agonists, prostacyclins/analogs, and endothelin pathway antagonists [49]. Seeger et al. noted that CPFE-related exercise limitation stems primarily from ventilatory dysfunction, rather than circulatory impairment [53]. Vasodilators risk worsening V/Q mismatch and hypoxaemia,targeted PH therapy is not routinely recommended. Current evidence supporting the survival benefit of targeted PH therapy in CPFE remains insufficient, with most studies being retrospective analyses and lacking high-quality prospective clinical trials for validation. Hyaluronan inhibition with 4MU attenuates fibrosis and PH in preclinical models, but clinical validation is needed [54].

Conclusion and outlook

CPFE is a distinct syndrome characterized by the simultaneous presence of emphysema and pulmonary fibrosis on HRCT, leading to preserved lung volumes but severely impaired gas exchange. Smoking is a key risk factor. Complications such as PH and lung cancer significantly worsen the prognosis. Current challenges include nonspecific manifestations, unclear pathogenesis, and the absence of effective targeted therapies. Enhanced understanding through prospective research is essential to refine diagnostic criteria, elucidate underlying mechanisms, and develop effective treatments to improve patient outcomes.

Key references

•[1] GLOBAL STRATEGY FOR THE DIAGNOSIS, MANAGEMENT, AND PREVENTION OF CHRONIC OBSTRUCTIVE PULMONARY DISEASE (2024 REPORT).

The report is the internationally authoritative guideline for COPD. Its importance is underscored by its role in:

Table 2 Evaluation of supporting evidence levels and limitations for major pharmacotherapeutic regimens in CPFE

Therapeutic Category	Representative Agent/Regimen	Mechanism of Action & Targeted Pathology	Primary Source of Evidence	Level of Evidence in CPFE & Key Findings	Major Limitations & Uncertainties
Antifibrotic Therapy	Nintedanib, Pirfenidone	Inhibits fibroblast proliferation and collagen deposition; targets pulmonary fibrosis progression	Large Phase III RCTs in IPF (e.g., INPULSIS, ASCEND)	Level IIb (Extrapolated Evidence): Observational studies and subgroup analyses suggest a potential to slow the rate of FVC decline in CPFE patients, though the efficacy may be weaker than in IPF alone	1. Evidence entirely extrapolated: No RCTs dedicated to CPFE exist 2. Heterogeneous efficacy: Emphysema may alter drug distribution or pathological response, rendering efficacy uncertain 3. Safety considerations: CPFE patients are typically elderly with multiple comorbidities, potentially exhibiting poorer tolerance to side effects (e.g., diarrhea with nintedanib, photosensitivity with pirfenidone)
Inhaled Corticosteroid/ Long-Acting β2-Agonist (ICS/LABA)	ICS/LABA combination inhalers	Anti-inflammatory and bronchodilatory; targets airflow limitation and airway inflammation	RCTs and epidemiological studies in COPD (e.g., TORCH, UPLIFT)	Level III (Low-quality Extrapolated Evidence): Small retrospective/prospective observational studies indicate potential improvement in FEV1 and reduction in acute exacerbations in some patients, but data are inconsistent	1. Weak and contradictory evidence: Based solely on small-scale, non-controlled studies 2. Unclear risk–benefit ratio: Long-term use increases pneumonia risk, requiring extreme caution in patients with concomitant fibrosis and recurrent infections 3. Unclear target population: Inability to predict which CPFE patients (e.g., those with predominant airway disease) might truly benefit
Targeted Therapy for Pulmonary Hypertension (PH)	PDE-5 inhibitors, Endothelin receptor antagonists, Prostacyclin analogues	Pulmonary vasodilation and anti-proliferation; targets pulmonary hypertension	RCTs in Group 1 PAH; exploratory studies in ILD-associated PH	Level III-IV (Very Limited & Contradictory Extrapolated Evidence): Mostly case series or retrospective analyses with conflicting results. Some suggest hemodynamic improvement without survival benefit, while others warn of potential worsening gas exchange	1. Lack of survival benefit evidence: No RCTs support improved prognosis in CPFE-PH 2. Potential risk of V/Q mismatch: Vasodilation may worsen ventilation/perfusion imbalance, potentially exacerbating hypoxemia—a key theoretical safety concern 3. Difficulty in patient selection: Inability to identify which CPFE-PH patients may safely benefit
Comprehensive Supportive Therapy	Long-term oxygen therapy, Pulmonary rehabilitation, Smoking cessation, Vaccination	Improves hypoxemia, enhances functional status, reduces triggers for acute exacerbations	High-quality RCTs and cohort studies in COPD and ILD populations	Level Ib-IIa (Direct & Strongly Extrapolated Evidence): Broad clinical consensus supports its foundational role. Observational studies confirm pulmonary rehabilitation improves exercise tolerance and quality of life in CPFE	Limitations pertain not to efficacy, but to implementation barriers: variable resource availability, patient adherence, and inconsistent execution in primary care settings

dynamic Providing an evidence-based, consensus framework for the definition, diagnosis, assessment, and management of COPD,serving as a fundamental reference for global clinical practice and research, guiding clinical decision-making, and promoting disease prevention.

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This review article provides a concise, integrative overview of pulmonary fibrosis, bridging fundamental pathogenesis with contemporary clinical practice. It provides a concise, authoritative update essential for both researchers and clinicians navigating diagnosis, treatment decisions, and patient-centered care.

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Smoking serves as a major risk factor for both emphysema and IPF, and is established as a primary etiology of CPFE, with a demonstrated dose–response relationship to cumulative smoking exposure. Distinguishing CPFE from other smoking-related interstitial lung diseases is therefore critical for accurate diagnosis and management. This perspective is supported by clinical analyses of smoking-related lung pathology, including Langerhans cell histiocytosis, as discussed in this literature. Progress in this area remains essential to address unresolved questions in disease mechanisms and clinical decisions.

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CPFE is increasingly recognized as a complex and distinct clinical syndrome. As highlighted in a comparative review of the condition, CPFE demonstrates unique clinical, physiological, and prognostic characteristics that differentiate it from either idiopathic pulmonary fibrosis or emphysema alone (Calaras et al., 2023). Research into CPFE is therefore vital to establish clear diagnostic criteria, understand its specific pathophysiology, and define optimal management strategies, moving beyond the approach used for its individual component diseases.

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The clinical management of CPFE remains challenging due to the absence of evidence-based therapies. Wang et al. (2022) explored the clinical efficacy of combined ICS/LABA inhalation in CPFE, underscoring the critical need for research to establish effective therapeutic strategies specifically and improve outcomes for this syndrome.

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This official ATS/ERS/JRS/ALAT statement provides the first consensus-based definitions for CPFE, establishing essential standardized criteria for diagnosis and research, which establishes a foundational consensus for standardizing research and improving clinical outcomes in CPFE. It consolidates current evidence and outlines future priorities, forming the foundational framework for the field.

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Tobacco smoking is a key etiologic driver of CPFE, a distinct clinical phenotype associated with accelerated functional decline and poorer outcomes. Research into CPFE is vital to delineate its unique pathophysiology from other smoking-related pathologies. Advancing this understanding is essential for accurate diagnosis and the development of targeted management strategies for this high-risk condition.

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CPFE represents a complex syndrome with a distinct clinical phenotype, yet its underlying molecular drivers remain poorly characterized. Recent research employing explainable artificial intelligence to analyze transcriptomic data from lung tissue has begun to elucidate the unique molecular landscape of CPFE, revealing interactive pathways distinct from IPF or COPD. This mechanistic distinction underscores the critical importance of studying CPFE as a separate entity. Deciphering its specific pathogenesis is fundamental for moving beyond descriptive diagnosis and towards the development of targeted biomarkers and therapies, ultimately enabling a more precise management approach for this high-mortality condition.

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Quantitative imaging is fundamental for characterizing the complex pathophysiology of CPFE. The application of advanced metrics, as outlined in joint society reports, is critical to objectively define and spatially map the coexisting patterns of fibrosis and emphysema that define this syndrome. Establishing such standardized quantitative criteria is essential to move beyond subjective visual assessment, enabling precise phenotype classification, monitoring of disease progression, and linking imaging features to functional impairment and clinical outcomes in CPFE.

••[31] Wells A U, Jacob J, Sverzellati N, et al. A formula for predicting emphysema extent in combined idiopathic pulmonary fibrosis and emphysema[J]. Respiratory Research, 2024, 25(1): 33.

Quantitative CT analysis has emerged as an objective tool for measuring emphysema and fibrosis extent. However, LAA%-quantitative CT metrics for emphysema- have limited ability to differentiate emphysema from fibrotic low-attenuation areas caused by traction bronchiectasis and honeycombing. Therefore, developing and validating more discriminative tools is critical. These advances are necessary to achieve accurate phenotyping, quantify the true burden of each component, and establish robust imaging-clinical correlations in this complex syndrome.

••[32] Ni H, Wei Y, Yang L, et al. An increased risk of pulmonary hypertension in patients with combined pulmonary fibrosis and emphysema: a meta-analysis[J]. BMC Pulmonary Medicine, 2023, 23(1): 221.

The incidence of pulmonary hypertension in patients with CPFE ranges from 15% tuo 55%, patients with CPFE carries a substantially worse prognosis.The pathogenesis

and potential pathophysiological mechanisms of CPFE with PH remain largely unclear.This emphasizes the need to warrant dedicated surveillance and management strategies to mitigate this major prognostic determinant.

••[37] Sadigov, A. S., et al. (2024). "MORTALITY RATE RELATED TO ACUTE EXACERBATIONS OF COMBINED PULMONARY FIBROSIS WITH EMPHYSEMA, IDIOPATHIC PULMONARY FIBROSIS, AND CHRONIC OBSTRUCTIVE PULMONARY DISEASE PATIENTS." CHEST 166(4, Supplement): A3178-A3179.

CPFE represents a distinct and understudied entity. The prognosis of acute exacerbation in CPFE is unclarity, conpared with acute exacerbations of PF,acute exacerbations of COPD which trend to be worst. This underscores the critical need for targeted research into CPFE's pathophysiology, risk stratification, and therapeutic strategies—gaps that must be addressed to improve clinical management and outcomes for this patient population. Feng X, Duan Y, Lv X, et al. The Impact of Lung Cancer in Patients with Combined Pulmonary Fibrosis and Emphysema (CPFE)[J]. Journal of Clinical Medicine, 2023, 12(3): 1100.

••[46] Zhao A, Gudmundsson E, Mogulkoc N, et al. Mortality surrogates in combined pulmonary fibrosis and emphysema[J]. European Respiratory Journal, 2024, 63(4).

CPFE remains a distinct, deeply understudied entity, whith nonspecific spirometric values,and not robust mortality surrogates. Zhao et al. (2024) address this gap by investigating mortality surrogates in CPFE, highlighting the pressing need for targeted research to refine prognostic assessment and advance therapeutic development for this patient population.

••[52] Glass D S, Grossfeld D, Renna H A, et al. Idiopathic pulmonary fibrosis: Current and future treatment[J]. The Clinical Respiratory Journal, 2022, 16(2): 84–96.

The management of pulmonary fibrosis in individuals with CPFE has largely been informed by clinical trials involving nintedanib and pirfenidone.CPFE, a distinct entity overlapping pulmonary fibrosis and emphysema, lacks tailored therapeutic guidelines despite advances in IPF management. This underscores the critical need for targeted research to bridge the gap between IPF-focused treatments and CPFE-specific clinical needs, optimizing outcomes for this understudied population.

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Authors' contributions
Luhong Chen: Conceptualization, Data curation, Writing—Original Draft. Xin Su: Literature collection, Data collection & organization. Xiulin Ye: Methodology design, Feasibility analysis, Critical revision of key sections. Xiaohua Du: Supervision, Project administration, Quality control & Final approval.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
This article does contain all reported studies/experiments with human or animal subjects performed by the authors have been previously published and complied with all applicable ethical standards(including the Helsinki declaration and its amendments, institutional/national research committee standards, and international/national/institutional guidelines).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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